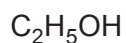


Answer **all** questions.

1. (a) The molecular formulae of five carbon compounds, **A**, **B**, **C**, **D** and **E** are shown below.

**A****B****C****D****E**

- (i) Alkanes are compounds with the general formula $\text{C}_n\text{H}_{2n+2}$.

Give the **letters** of the **two** compounds that are alkanes.

[1]

..... and

- (ii) Give the **name** for compound **A**.

[1]

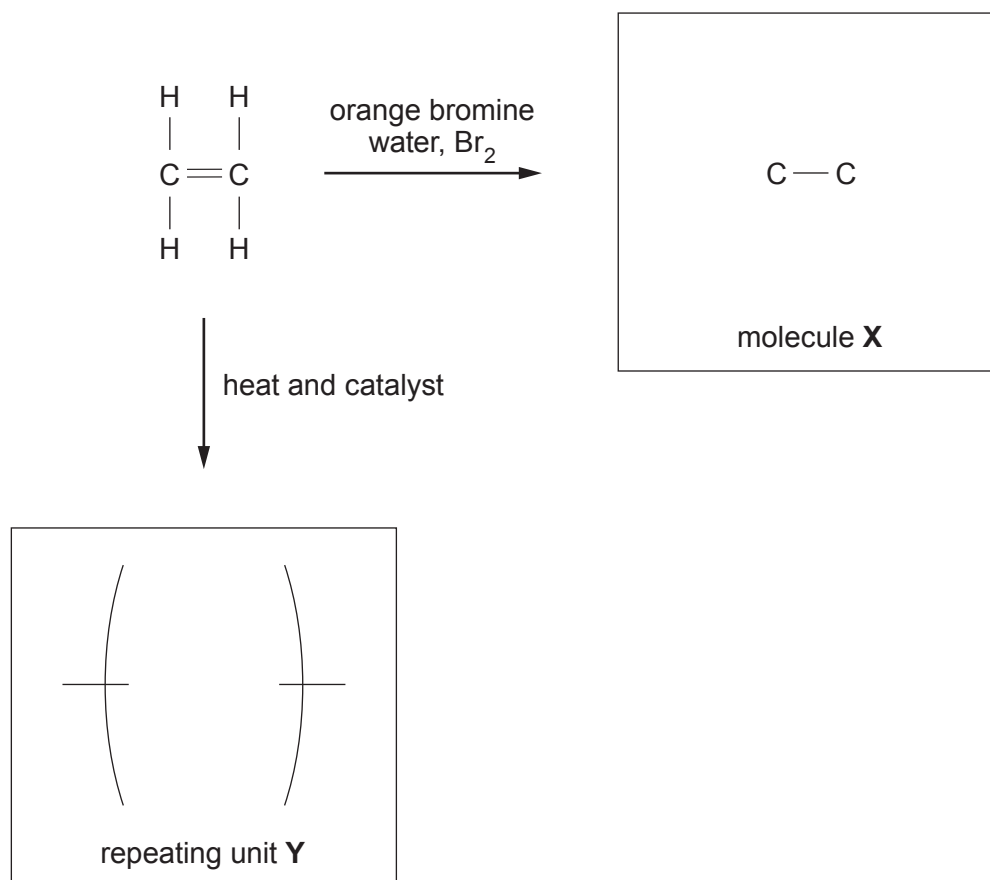
.....

- (iii) Draw the structural formula of compound **E**.

[1]



(b) The following diagram shows two reactions of ethene.



(i) **Complete the structural formulae** for molecule **X** and repeating unit **Y**. [2]

(ii) Describe the change that would be seen in the orange bromine water when molecule **X** is formed. [1]

(iii) Name the type of reaction that takes place to form repeating unit **Y**. [1]



(c) Paper and plastic are both used to make disposable shopping bags.



The table compares the percentages of disposable paper and plastic bags used by consumers in a Welsh town between 1988 and 2018.

Type of bag	Percentage of disposable bags used (%)						
	1988	1993	1998	2003	2008	2013	2018
paper	73	63	31	12	8	13	21
plastic	27	37	69	88	92	87	79

(i) In 2003, the consumers of the town used a total of 1.25 million disposable bags.

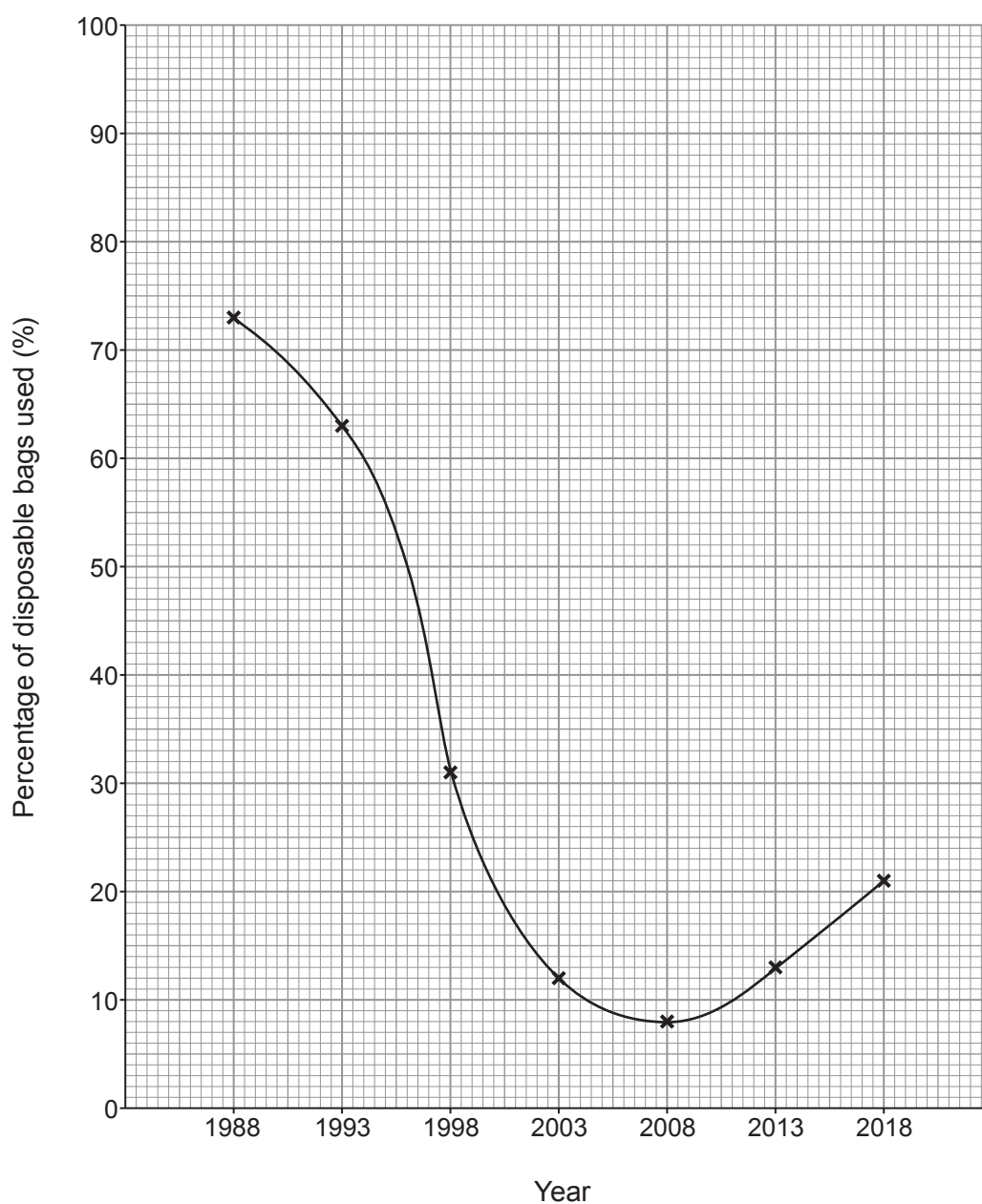
Use the information in the table to calculate the number of **paper** bags used. [2]

Number of paper bags = million



- (ii) The graph below shows the percentage of **paper** bags used between 1988 and 2018.

Plot on the **same grid** the percentage of **plastic** bags used between 1988 and 2018. Draw a suitable line. [3]



(iii) **Use the graph** to answer the following questions.

- I. Give the year when **equal** numbers of plastic bags and paper bags were used. [1]

Year

- II. Circle the **simplest** ratio that compares the use of paper bags to plastic bags in 2000. [1]

2 : 8

1 : 4

4 : 1

8 : 2

20 : 80

- (iv) Give **one** reason that explains the change seen in plastic bag usage in Wales between 2008 and 2018. [1]

.....

.....

